

## Numerical Calculation: Problems

1. **Computing Expressions.** Consider the golden ratio  $\phi = \frac{1+\sqrt{5}}{2}$ .

- a) Test that  $\phi^2 - \phi - 1 = 0$ .
- b) Compute  $(\sqrt[6]{\pi^5 + \pi^7})^{4/\sqrt{\phi}}$ , show 10 significant digits.

2. **Symbolic Computations.**

- a) Demonstrati folosind calcul simbolic identitatea urmatoare

$$(a_1^2 + a_2^2)(b_1^2 + b_2^2) = (a_1b_1 + a_2b_2)^2 + (a_1b_2 - a_2b_1)^2.$$

- b) Factorizati  $50!$  si gasiti numarul de zero-uri al rezultatului.

3. **Derivatives and primitives**

- a) Compute the first and second derivatives of  $f(x) = \frac{\sin(x^2 + 1)}{\sqrt{x^4 - x^2 + 1}}$
- b) Compute the primitive of  $e^{3x} \sin x$
- c) Compute the limit  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n+n^2}\right)^{3n^2}$

4. **Computing definite integral and plotting**

- a) Compute  $\int_{-1}^1 \sqrt{1-x^2}$
- b) Plot the region between the function and the Ox axis.

5. **Solve an equation**

- a) Solve the equation  $x^3 - x + 1 = 0$ , finding all the roots.
- b) Solve the equation  $x^3 - x + 1 = 0$  using a solve block starting from the initial condition  $x = -1$ . What is the root found? Is it among the roots presented at point a)? What happens if we change the initial condition to  $x = 1$ ?